TSA 2013

Problem Solving

Questions taken from the TSA Oxford 2013 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2013 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) Three thermometers are each accurate to within 2 degrees above or below the temperature they actually read. One reads 7°, one reads 9° and one reads 10°.

What is the minimum range in which the true temperature lies?

A 5°–12°

B 7°–9°

C 8°–10°

D 8°–9°

E 7°–10°

2 (Q6) On only 360 occasions during the twentieth century was it possible to write the date in the form DD/MM/YYYY using eight different digits. One example was 28th July 1956 (28/07/1956).

Eleven of Arthur's birthdays have been such dates (though not his date of birth), and in 1974 and 1983 the two digits that did not form part of the date made up his age at the time.

In which year was Arthur born?

A 1909

B 1916

C 1918

D 1936

E 1938

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3 (Q7) In order to qualify for a bonus, employees must fulfil certain criteria:

£1,000 bonus	Absences less than 5%
	Production target exceeded by at least 10%
	Rejects are less than 5% of output
£500 bonus	Absences less than 10%
	Production targets met
	Rejects are less than 8% of output

Workers performed as follows:

	Smith	Jones	Patel	Owololu	McKay
Attendance (%)	95	90	100	96	97
Over production target (%)	+5	+6	+12	0	-4
Product accepted (%)	98	96	95	93	96

Who qualifies for a bonus?

A Nobody

B Smith

C Patel and Smith

D Owololu, Patel and Smith

E Jones, Owololu, Patel and Smith

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4 (Q8) I stopped in a town called Akeland to ask the way to Eksberg. The only sense I could get from a local was as follows:

Akeland is east of Benford.

Eksberg is north of Cranton.

Ducton is south of Akeland.

Cranton is west of Ducton.

Assuming that 'east' means, as near as it matters, 'due east', which one of the following can be said with certainty?

A I need to go due west.

B I need to go to the west, but it may be north-west or south-west.

C I need to go north-west.

D I need to go to the north, but it may also be a bit west.

E I need to go south-west.

5 (Q12) I have just joined a savings scheme. I will pay in £50 each month and at the end of each complete year I will be paid 5% interest on my average balance (i.e. the average of my starting and finishing amount) for the previous 12 months.

How much will I have in total when my interest has been paid at the end of the second year? (Answer to the nearest £10.)

A £1230

B £1260

C £1290

D £1380

E £1990

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6 (Q13) Every Thursday, Marilyn spends 10 hours making 60 cakes to sell at her local market. The cost of the ingredients to make each cake is £1.60 and Marilyn charges £6 an hour for her time. She usually sets the sale price for a cake at 75% more than the total costs of making it. However, when she sells to her friends she gives them a 10% discount on the normal sale price.

What do her friends have to pay for one of Marilyn's cakes?

A £3.42

B £4.10

C £4.29

D £4.55

E £4.88

7 (Q14) A business, facing turbulent times, changed the wages of its workers by the same percentage in two successive years, but the changes were in opposite directions.

Which of the following best describes the wages of the workers after the second change?

A The same as the original wage.

B Necessarily higher than the original wage.

C Necessarily lower than the original wage.

D Higher than originally if the first change was an increase, otherwise lower.

E Higher than originally if the first change was a decrease, otherwise lower.

8 (Q18) Sven is a keen cyclist and is riding from Alvros to Berget. At his normal cycling speed this will take him 30 minutes. His wife, Helga, will go by car to bring him and his bike back. They leave at the same time, but two thirds of the way between the two towns the car breaks down and Helga has to walk the rest of the way. The car goes at three times the speed Sven cycles. But Helga walks at only one third of the speed he cycles.

Who arrives at Berget first, and by how much (to the nearest minute if necessary)?

A Helga by 12 minutes.

B They arrive at the same time.

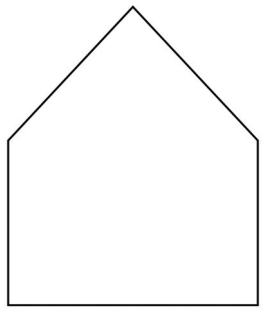
C Sven by 7 minutes.

D Sven by 33 minutes.

E Sven by 37 minutes.

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9 (Q19) I am making a patchwork quilt from leftover pieces of cloth of various shapes and sizes. This pentagonal piece of cloth is rather large, so I want to cut it into two pieces.



Which one of the following combinations is it **not** possible to produce by making one straight cut across a pentagonal piece of cloth?

A A triangle and a quadrilateral.

B A triangle and a pentagon.

C A triangle and a hexagon.

D A quadrilateral and a pentagon.

E A quadrilateral and a hexagon.

10 (Q20) Thomas needs to book a place to sit a test next week, but is free only in the mornings. The test centre runs the test every day from Monday to Friday and holds a morning and an afternoon sitting. Each sitting has a total of 100 places available and will take place in the main hall unless there are 30 or fewer candidates, in which case a smaller room is used. He does not like to take tests in crowded places, so he wants to make sure that he is in the main hall and will try to be with the smallest number of other candidates possible. The number of places that are still available are shown in the table below.

Day	Spaces available a.m.	Spaces available p.m.
Monday	75	67
Tuesday	48	47
Wednesday	57	2
Thursday	39	10
Friday	63	46

On which day should he choose to sit the test?

A Monday

B Tuesday

C Wednesday

D Thursday

E Friday

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11 (Q24) Mr Daley has bought some used carpeting for his new 24 m x 12 m car showroom. The carpet is in 8 m x 4 m rectangles which will be joined using a strip of double-sided tape along all seams. Additionally, double-sided tape will be needed to stick the carpet around the edges of the showroom.

What is the minimum amount of tape he requires?

A 72 m

B 108 m

C 144 m

D 216 m

E 288 m

12 (Q25)

WINTER SPECTACULAR

AT

THE PLAYTIME THEATRE

From: Saturday 7th December

To: Saturday 28th December

Performances nightly at 7 pm (Sun–Fri)

(Saturdays at 5.30 pm and 8.30 pm)

Matinees Weds, Thurs and Sat at 2 pm

Tickets from £8

(Theatre closed 25th December)

How many times will the Winter Spectacular be performed at the Playtime Theatre during its run?

A 31

B 34

C 35

D 36

E 38

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13 (Q26) A popular option at the Pizza Parlour is the 'Plus Two' pizza, which is a basic cheese and tomato pizza plus a choice of any two different additional toppings from the following list:

Tuna Ham Chicken Onion Mushroom Sweetcorn

This gives 15 variations, which can be calculated as follows:

Each of the six additional toppings can be combined with one of the other five to create 30 pairs, but this counts each pair twice, e.g. tuna and ham; ham and tuna.

Which of the following does not give a result of 15?

A The total number of matches when six teams play each other once.

B The total number of handshakes when six people shake hands with each other.

C The total number of crackers needed for six children to pull a cracker with each of the others.

D The total number of partnerships possible when two people are chosen from a shortlist of six to represent their society in a debate.

E The total number of cards sent when six friends send each other Christmas cards.

14 (Q30) An acre of land planted with sugar beet produces 550 gallons of ethanol from the sugar by fermentation. A car converted to run on E75 (a 75% ethanol, 25% petrol mixture) can do 40 miles per gallon.

If you drive 20,000 miles per year, how many acres of sugar beet will be needed to produce the ethanol you need?

A (20000/40 x 25/100)/550 = 0.23

B (20000/40 x 75/100)/550 = 0.68

C (20000/40 x 100/75)/550 = 1.21

D (40/20000 x 100/75) x 550 = 1.47

E (20000/40 x 100/25)/550 = 3.63

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15 (Q31) The following table gives information about five different car models.

Model	Engine capacity (litres)	Fuel tank capacity (litres)	Maximum fuel economy (miles per litre)
Clipper	1.5	60	12
Ghia	2.0	70	11
Sedan	2.5	75	10
Estate	3.0	80	8
Saloon	4.0	82	5

Which of these models will travel furthest on one full tank of petrol under optimum conditions?

A Clipper

B Ghia

C Sedan

D Estate

E Saloon

16 (Q32) The table below shows the percentage of adults, grouped by age and socio-economic group, with no natural teeth.

	Age group					
	25–34	*35–44*	*45–54*	*55–64*	*65–74*	*75+*
Professionals, employers and managers	1	3	10	22	48	71
Intermediate	1	4	14	34	53	71
Skilled manual	2	7	23	48	66	82
Semi-skilled and unskilled manual	3	14	34	58	71	87

Which one of the following conclusions can be drawn from the table above?

A Most people have lost their natural teeth by the age of 55.

B Professional people tend to keep their natural teeth longer than semi-skilled and unskilled manual workers.

C Manual work is bad for your teeth.

D People without natural teeth tend not to be employed as managers.

E There are over three times as many 45–54 year-old manual workers with no teeth as professionals.

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17 (Q36) Last month, in an attempt to improve my fitness, I began to run daily. The maximum distance I have time to fit in on any particular day is 8 miles, but my aim is to maintain an average of at least 5 miles per day.

After 15 days my average was just under 6 miles per day, then the following day I increased my average to just over 6. Since then (for the last 9 days, up to and including today) I have been able to run only 2 miles each day.

What is the **minimum** number of days from now until I can bring my average back up to 5 miles per day?

A 2 days

B 4 days

C 9 days

D 18 days

E 26 days

18 (Q37) I have a bird automaton that, when activated, flaps its wings and whistles three different tunes successively, selected at random from its repertoire of ten. It performs one of its ten tunes for 10 seconds, six of them for 15 seconds each and three of them for 20 seconds each.

When my granddaughter visits she is fascinated, activating it again and again as soon as it stops each time. On her last visit she kept it going for a whole hour, and I lost count of how many times she activated it.

What is the **maximum** number of times that my automaton could possibly have been activated during one hour?

A 75

B 80

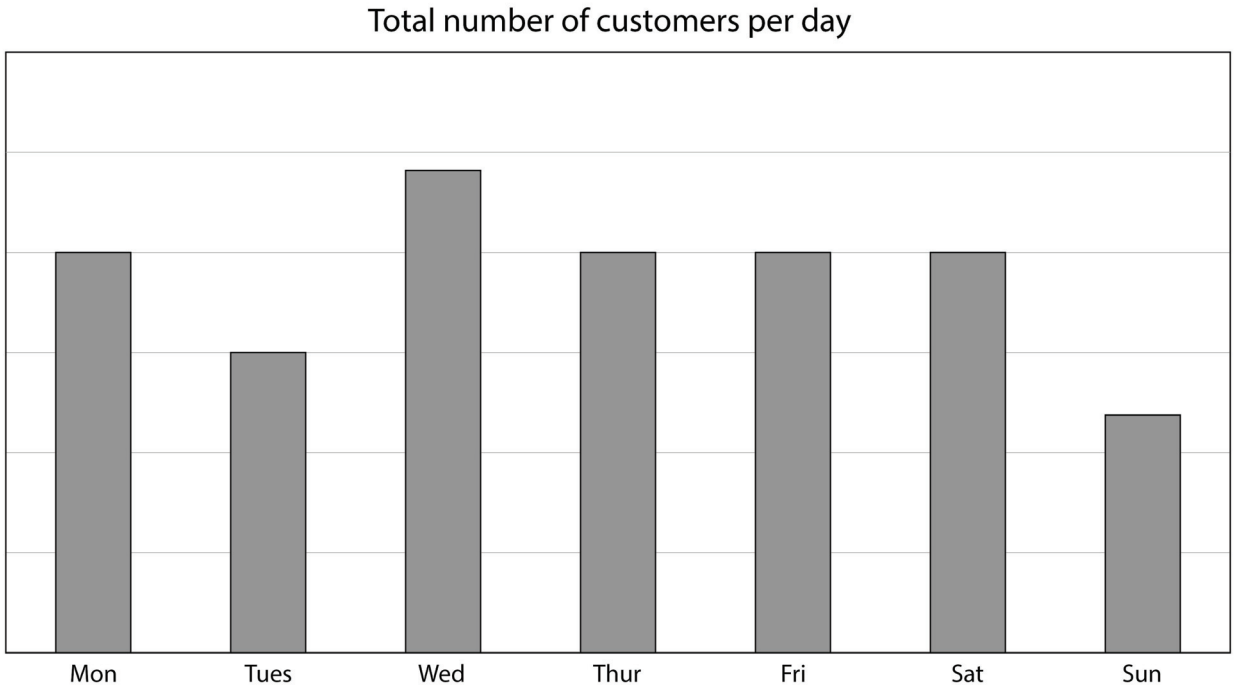
C 90

D 120

E 225

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19 (Q38) Graham recorded the number of visitors to his shop each day last week and presented the results in the bar chart below:



When he calculated the number of customers per hour he found that he had the same number for five of the days, but the values for Tuesday and Friday were slightly higher. The opening hours of the shop are as follows:

Day	Open	Close
Monday	8 am	6 pm
Tuesday	?	?
Wednesday	8 am	8 pm
Thursday	8 am	6 pm
Friday	?	?
Saturday	8 am	6 pm
Sunday	10 am	4 pm

Which one of the following could be the opening hours of the shop for Tuesday and Friday?

A Tuesday 9.30 am–5.30 pm, Friday 8.00 am–6.30 pm

B Tuesday 9.00 am–4.30 pm, Friday 8.30 am–6.00 pm

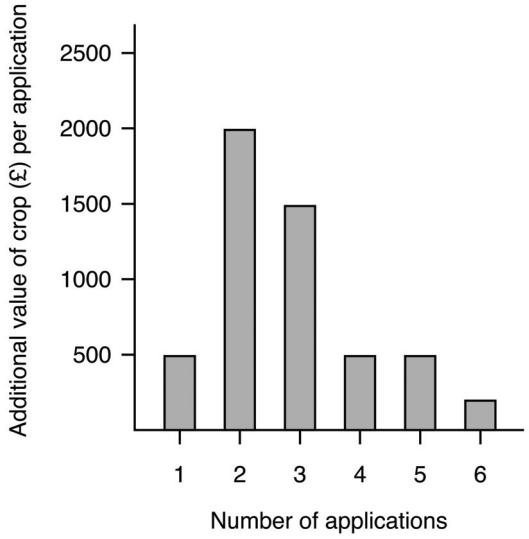
C Tuesday 9.00 am–4.30 pm, Friday 8.00 am–6.00 pm

D Tuesday 9.00 am–5.00 pm, Friday 8.00 am–6.30 pm

E Tuesday 10.00 am–5.00 pm, Friday 8.30 am–6.00 pm

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20 (Q42) The graph below shows the additional financial value of repeated applications of fertiliser applied to crops in a field for a year. For example, two applications gives £2,000 more value to the crop than one application.



If each application of fertiliser costs £1,000, how many applications should be made to maximise the farmer's profit?

A 0

B 2

C 3

D 4

E 5

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21 (Q43) There is a half-hourly train service that operates along the Rowbina Valley, between Erd and Teovil. Leaving Erd at 10 and 40 minutes past each hour, trains stop at Aregon, Lowley, Ergen, Uble and Gidion before arriving at Teovil 51 minutes after departure from Erd. In the opposite direction, trains leave Teovil at 20 and 50 minutes past each hour, arriving at Erd 51 minutes later.

Distances Chart (km)

Erd	Aregon	Lowley	Ergen	Uble	Gidion	Teovil
6						
11	5					
17	11	6				
24	18	13	7			
29	23	18	12	5		
36	30	25	19	12	7	

The trains normally average a speed of 60 km/h, but this rises to 80 km/h if they are running late. They stop for three minutes at each station en route, but this is reduced to two minutes when running late.

Yesterday the 10.20 am train from Teovil arrived at Uble on time, but because of a technical fault, remained at Uble for 22 minutes. There were no further delays.

How many minutes late was yesterday's 10.20 am train from Teovil when it arrived at Erd?

A 5 mins

B 8 mins

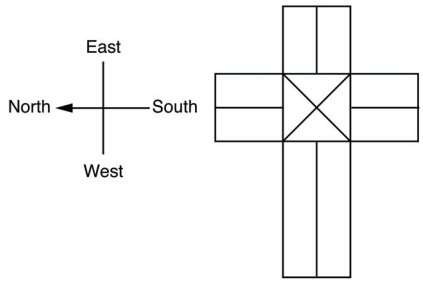
C 10 mins

D 13 mins

E 15 mins

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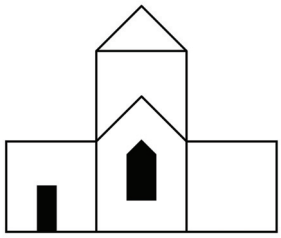
22 (Q44) The diagram below shows a view of a church from above:



There are three windows facing south, two windows facing north and a stained glass window facing east. The only doors are one in the westernmost end and one in a north-facing wall.

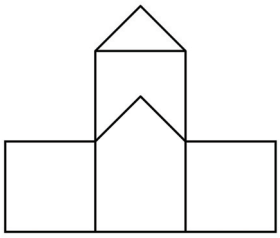
Which one of the following gives a possible view of the church?

A



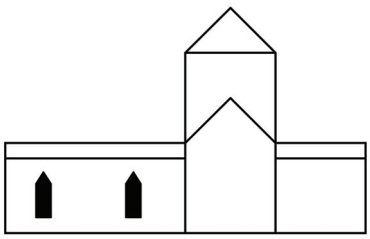
View from East

B



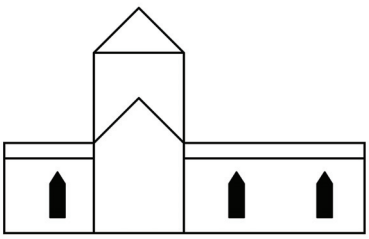
View from West

C



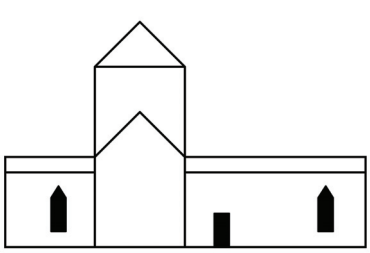
View from South

D



View from South

E



View from North

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23 (Q48) When travelling in Kromistan and not understanding the currency, I offered a red note for an item marked '135K'. I was given, in change, three green coins and one blue coin. Later, for a newspaper marked '33K', I offered a handful of coins and the vendor took four green and one blue. There are only these three denominations of money available; the smallest is marked '1K' and each higher denomination is a whole number multiple of the lower denominations.

How many green coins are worth one red note?

A 8

B 20

C 29

D 160

E 167

24 (Q49) At committee meetings of the Massing Social Club motions are carried or defeated by a simple majority, abstentions being ignored.

At the last meeting a proposal to install a satellite dish was defeated by 8 votes to 5 with 10 committee members abstaining. However, because of the large number of abstentions it was decided that the matter should be discussed further at the next meeting and a vote taken again.

What is the minimum number of committee members who must change the way they voted if the result is to be reversed at the next meeting?

A 2

B 4

C 7

D 8

E 12

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25 (Q50) The housing estate on which Andrew lives has streets of variable length. In each street the houses are directly opposite each other, with the same number of houses on each side. However, the numbering of the houses is inconsistent.

Some streets are numbered consecutively on each side, starting at one end of the street, going up to the end and then back down the other side.

Other streets have the odd numbers on one side, starting from 1, and the evens on the other side, starting with 2 at the same end.

Andrew's friend Amy lives on the same estate at 25 Acacia Avenue, which is not the end house in the street.

If Andrew knows the number of the house opposite Amy's, which one of the following statements is definitely true?

A He knows the exact number of houses in the street.

B He knows that the number of houses in the street is one of two numbers.

C If the house opposite Amy's is 26, he knows the number of houses in the street.

D If he knows the number of the house next door to Amy, he can work out the number of houses in the street.

E He knows the numbers of the houses immediately adjoining Amy's.

END OF TEST

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